MINI NLP PROJECT REPORT

SmartResume: An NLP-Based Resume–Job Matching and Skill Gap Analysis System

Research Question

How do traditional TF-IDF similarity and modern semantic embeddings differ when matching resumes with job descriptions?

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Abstract

Resume screening involves comparing the information in a candidate's resume with the requirements of a job description. A simple keyword-based comparison can miss cases where two texts express similar ideas using different words. This project develops SmartResume, a small NLP-based system that compares resumes and job descriptions using two complementary approaches. The first is TF-IDF with cosine similarity, which acts as a traditional lexical baseline. The second uses the pretrained all-MiniLM-L6-v2 Sentence Transformer to obtain semantic embeddings and measure meaning-level similarity. A hybrid score combines the two approaches, while a rule-based skill-gap component identifies skills that are present or missing. The system was tested on 2,385 resume–job pairs from the selected dataset. In the controlled evaluation, TF-IDF achieved a ROC-AUC of 0.853 and semantic similarity achieved 0.843. The results show that the two approaches provide different but related signals, with semantic embeddings often producing higher similarity when the wording differs. The project also demonstrates how a simple, explainable NLP pipeline can be extended with a modern pretrained language model.

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1. Introduction

Recruitment systems often need to compare a large number of resumes with job descriptions. The relevant information is mainly available as unstructured text, which makes the problem suitable for Natural Language Processing (NLP). Traditional text-matching methods can identify shared words, but they may not fully capture the meaning of related terms or phrases.

SmartResume was developed as a small practical NLP application to study this difference. The project compares a traditional TF-IDF and cosine-similarity approach with semantic similarity obtained from a pretrained Sentence Transformer. The project also adds a skill-gap analysis layer so that the output is not limited to a single similarity score.

The main research question is: How do traditional TF-IDF similarity and modern semantic embeddings differ when matching resumes with job descriptions? The comparison is useful because it shows the trade-off between lexical overlap, which is easy to understand, and semantic representations, which can capture related meaning even when exact words do not match.

2. Problem Statement

The objective of this project is to develop an NLP-based system that can compare a candidate resume with a job description and provide an interpretable estimate of how well the resume matches the job. The system should also identify required skills that are present in the resume and highlight skills that appear to be missing.

The problem is an NLP problem because both resumes and job descriptions are text documents. The system must preprocess the text, represent it numerically, measure similarity between the two documents, and extract relevant skill information.

3. Objectives

- Preprocess resume and job-description text for similarity analysis.
- Measure lexical similarity using TF-IDF and cosine similarity.
- Measure semantic similarity using a pretrained Sentence Transformer.
- Compare the traditional and semantic similarity approaches.
- Identify matching and missing skills using rule-based skill-gap analysis.
- Combine the two similarity scores into a hybrid match score.
- Classify the result into Strong Match, Moderate Match, or Low Match.
- Evaluate the approaches using genuine and deliberately mismatched resume–job pairs.
- Demonstrate the final SmartResume analyzer on a new resume and job description.

4. Dataset

The project uses the job_resume_fit.csv dataset available through Hugging Face. The dataset contains resume–job-description pairs together with job categories, required skills, candidate skills, and several existing matching scores.

Dataset Property	Details
Dataset name	job_resume_fit.csv
Source	Hugging Face Datasets
Number of records	2,385
Number of columns	10
Job categories	23
Text fields	resume_text, job_text
Skill fields	job_required_skills, resume_skill_list, ai_matched_skills
Other score fields	ai_match_score, skill_string_match_score, fuzzy_match_score
Target variable	No independently verified relevance label is provided

The dataset has no missing values in the fields used by the project. The job categories include domains such as HR, CHEF, FINANCE, AVIATION, BANKING, INFORMATION-TECHNOLOGY, HEALTHCARE, ENGINEERING and others.

A limitation of the dataset is that its ai_match_score is already supplied by the dataset and is not independently verified. Therefore, it is treated only as an auxiliary/reference score and not as ground truth.

5. Methodology

The overall workflow of SmartResume is:

Resume + Job Description	Preprocessing	TF-IDF + Cosine	Sentence Transformer	Hybrid Score	Skill Gap Analysis	Final Match + Recommendations

5.1 Text Preprocessing

The preprocessing step is intentionally simple and explainable. Text is converted to lowercase, punctuation and special characters are removed, and repeated whitespace is normalized. The cleaned resume and job-description text are then used for similarity calculations.

5.2 TF-IDF and Cosine Similarity

TF-IDF (Term Frequency–Inverse Document Frequency) represents a document using the importance of its words relative to the document collection. Cosine similarity is calculated between the TF-IDF vectors of the resume and job description. This provides a transparent lexical baseline: a higher score generally indicates greater overlap in the vocabulary used by the two texts.

5.3 Semantic Similarity using Sentence Transformers

The project uses the pretrained all-MiniLM-L6-v2 Sentence Transformer. Instead of relying only on exact word overlap, the model converts the resume and job description into 384-dimensional embeddings. Cosine similarity between these embeddings provides a semantic similarity score. No supervised training is performed; the pretrained model is used only to generate embeddings.

5.4 Skill Gap Analysis

The dataset provides required skills for the job and a list of skills associated with the candidate. These lists are parsed and compared using set intersection and set difference. Matching skills are required skills that are also present in the candidate list, while missing skills are required skills absent from the candidate list.

5.5 Hybrid Matching

The final hybrid score is calculated as:

$$\textbf{Hybrid Score} = 0.4 \times \textbf{TF-IDF Score} + 0.6 \times \textbf{Semantic Score}$$

The semantic component receives a larger weight because it captures contextual similarity, while TF-IDF provides a transparent lexical signal. These weights are heuristic design choices for this mini-project and were not learned from labelled training data.

5.6 Match Classification

Hybrid Score	**Classification**
≥ 70%	Strong Match
45% – < 70%	Moderate Match
< 45%	Low Match

These thresholds are heuristic and are used to provide a simple human-readable category rather than a clinically or statistically calibrated prediction.

6. Implementation

The project was implemented in Python using Google Colab. The main libraries used were Pandas and NumPy for data handling, re and ast for text cleaning and skill-list parsing, Matplotlib for visualization, scikit-learn for TF-IDF, cosine similarity and ROC-AUC, and Sentence Transformers for semantic embeddings.

Component	**Technology / Method**
Programming language	Python
Environment	Google Colab
Data handling	Pandas, NumPy
Text preprocessing	Regular expressions
Traditional NLP	TF-IDF + cosine similarity
Semantic NLP	Sentence Transformer: all-MiniLM-L6-v2
Evaluation	ROC-AUC and controlled pairwise comparison

Visualization	Matplotlib
Skill extraction	Rule-based phrase matching

The final SmartResume function accepts a resume and a job description, cleans both texts, computes the two similarity scores, calculates the hybrid score, performs skill-gap analysis, classifies the match, and generates a short recommendation based on missing skills.

7. Results and Discussion

The results below use the values produced by the Colab notebook. The assignment requires appropriate metrics, tables, graphs, sample outputs and screenshots where applicable; therefore, the main experimental graphs and a final sample output are included in this section.

7.1 TF-IDF Similarity Results

Statistic	TF-IDF Match Score
Average	16.03%
Median	12.79%
Minimum	0.00%
Maximum	59.88%

```
TF-IDF Match Score Statistics
----------------------------
Average: 16.03%
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```

Figure 1. Figure 1. TF-IDF match score statistics.

The distribution is concentrated toward lower TF-IDF scores, which is expected from a lexical method when resumes and job descriptions do not share many exact terms. The maximum observed score was 59.88%, while the average was 16.03%.

7.2 TF-IDF versus Semantic Similarity

The average TF-IDF similarity was 16.03%, whereas the average semantic similarity was 51.93%. The correlation between the two scores was 0.667, showing that they are positively related but are not measuring exactly the same signal.

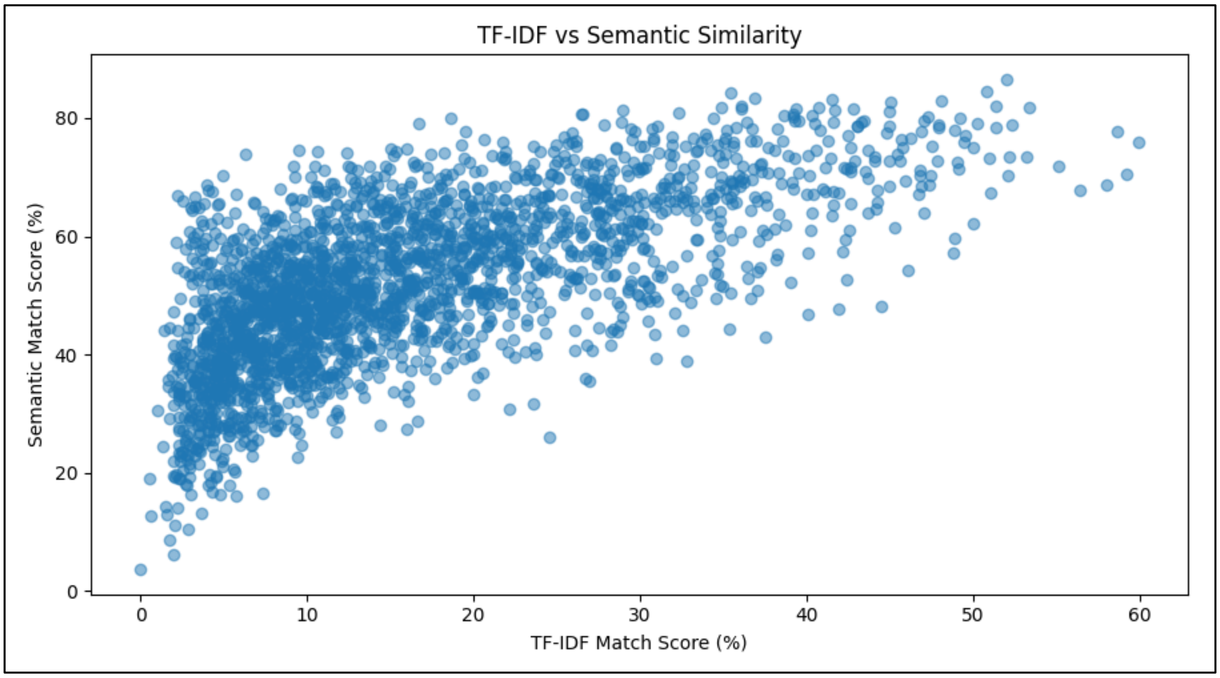

Figure 2. TF-IDF versus semantic similarity for the resume–job pairs.

The scatter plot shows a general upward relationship, but there is substantial spread. This supports the project's main comparison: semantic embeddings can assign higher similarity when the underlying meaning is related even when exact lexical overlap is limited.

7.3 Genuine versus Controlled Mismatched Pairs

Because the dataset does not contain independently verified human relevance labels, the project does not claim conventional classification accuracy. Instead, a controlled pairwise experiment was performed. The genuine pair is the original resume with its associated job description. For each resume, a negative pair was constructed by selecting a job description from a different job category. This produces 2,385 genuine pairs and 2,385 controlled mismatched pairs, for 4,770 evaluation pairs in total.

Measure	Genuine pairs	Different-category mismatches
Average TF-IDF similarity	16.03%	5.24%
Average semantic similarity	51.93%	34.45%

Both methods produced lower average similarity for the controlled mismatches. The reduction was from 16.03% to 5.24% for TF-IDF and from 51.93% to 34.45% for semantic similarity.

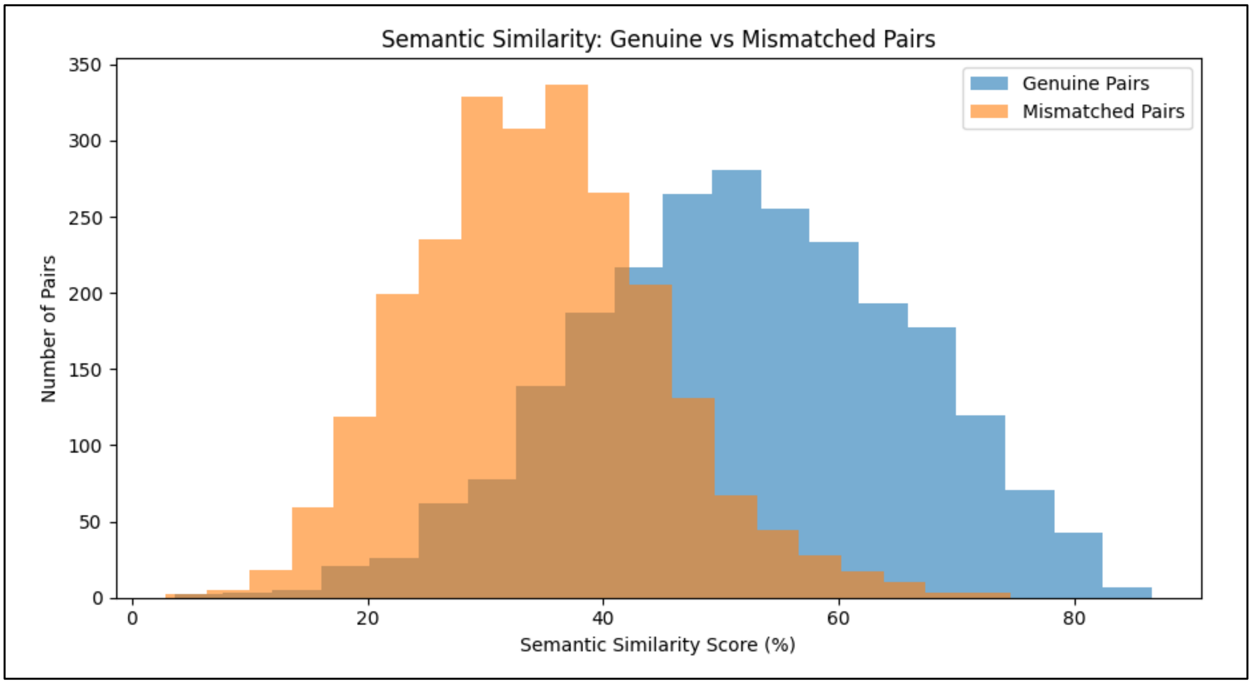

Figure 3. Semantic similarity distribution for genuine and controlled mismatched pairs.

Method	ROC-AUC
TF-IDF	0.853
Semantic similarity	0.843

The ROC-AUC values indicate that both similarity signals can distinguish the genuine pairs from the deliberately constructed different-category pairs in this controlled experiment. TF-IDF produced a slightly higher ROC-AUC than semantic similarity in this particular evaluation.

7.4 Match Category Distribution

Using the heuristic hybrid thresholds, most dataset pairs fall into the Low Match category. There were 1,777 Low Match pairs, 604 Moderate Match pairs and 4 Strong Match pairs.

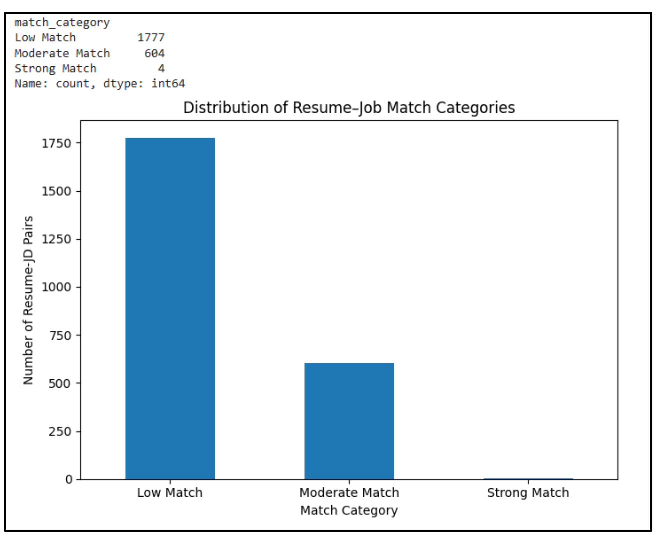

Figure 4. Distribution of resume–job match categories.

7.5 Category-Level Analysis

Average scores were also examined across job categories. The category-level analysis showed that CHEF had the highest average hybrid score at about 51.49%, followed by HR at about 48.60% and CONSTRUCTION at about 46.61%. The lowest average hybrid score was observed for ARTS at about 21.38%.

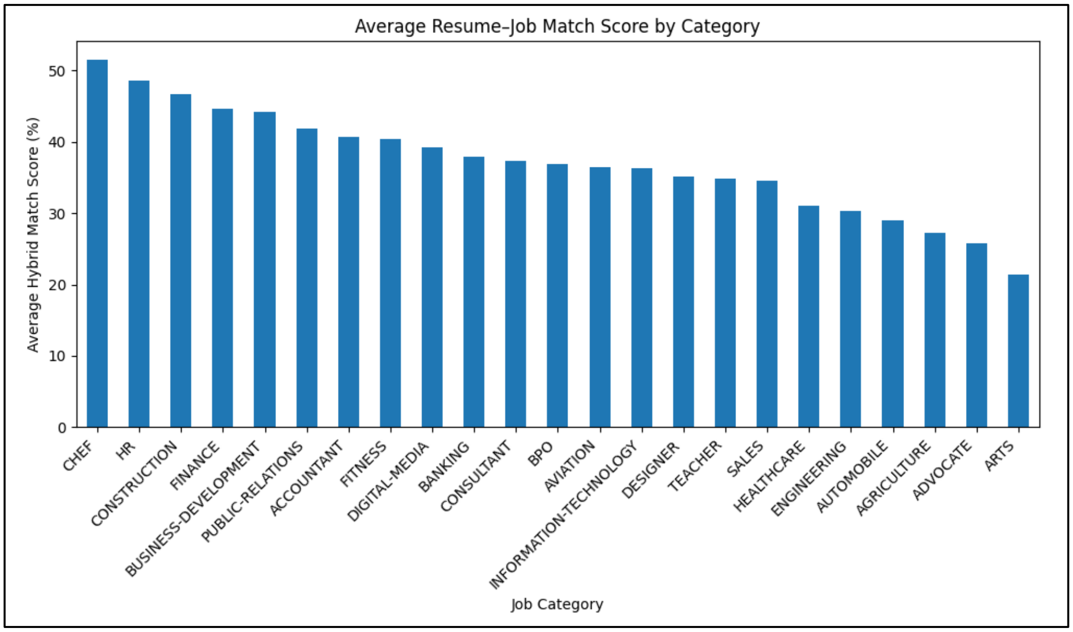

Figure 5. Average hybrid resume–job match score by job category.

7.6 Skill Gap Analysis

Across the dataset, the average skill-match percentage was 11.55%, with a highest observed value of 41.67% and a lowest value of 0.00%. The relatively low average is partly a result of the strict set-based comparison: a skill must be represented in the selected skill lists to count as a match.

7.7 Interesting Case Study

One useful example was identified in the AVIATION category. For this pair, TF-IDF similarity was only 6.27%, while semantic similarity was 73.99%, a difference of 67.72 percentage points. This example illustrates why semantic embeddings can be useful when two texts discuss related concepts without sharing a large amount of exact vocabulary.

7.8 Final SmartResume Sample Output

The final application was demonstrated using a new sample resume describing a BCA student with experience in Python, SQL, Pandas, NumPy, scikit-learn, machine learning and NLP, and a junior data-analyst job description requiring those skills along with data preprocessing, data visualization and statistical analysis.

```
SMARTRESUME ANALYSIS
====================
TF-IDF Score: 62.52%
Semantic Score: 78.22000122070312%
Hybrid Match Score: 71.94%
Overall Result: Strong Match
Skill Match: 87.5%

Matching Skills:
✓ data preprocessing
✓ data visualization
✓ machine learning
✓ nlp
✓ pandas
✓ python
✓ sql

Missing Skills:
✗ statistical analysis

Recommendations:
- Consider improving or gaining experience in: statistical analysis.
```

Figure 6. Final SmartResume sample output from the corrected application run.

The corrected final output gives a TF-IDF score of 62.52%, semantic score of 78.22% and hybrid score of 71.94%, resulting in a Strong Match classification. The skill-match percentage is 87.5%. Seven required skills are identified as matching, while statistical analysis is identified as the remaining skill gap.

7.9 Main Findings

TF-IDF provides a simple and interpretable lexical baseline.

Semantic embeddings generally produce higher similarity because they capture related meaning beyond exact word overlap.

The two similarity methods are positively correlated but not identical (correlation ≈ 0.667).

In the controlled evaluation, TF-IDF achieved ROC-AUC 0.853 and semantic similarity achieved 0.843.

The hybrid score combines lexical and semantic information but uses heuristic weights.

Skill-gap analysis makes the final result more interpretable by showing matching and missing skills.

8. Limitations

- The dataset does not provide independently verified human relevance labels, so the evaluation uses controlled different-category mismatches rather than human-judged relevance.
- The hybrid weights of 40% TF-IDF and 60% semantic similarity are heuristic and were not learned from labelled data.
- The Strong/Moderate/Low classification thresholds are heuristic.
- Skill extraction is rule-based and depends on the selected vocabulary. Synonyms and equivalent skill phrases may not always be detected.
- Sentence Transformer inference is computationally heavier and slower than TF-IDF.

- The system does not assess the quality of a candidate's experience, seniority, achievements or context beyond the supplied text.
- The final score should be treated as a decision-support signal rather than a replacement for human recruitment judgment.

9. Future Scope

- Collect human-verified relevance labels and use them to learn or calibrate the hybrid weights.
- Experiment with additional transformer models or domain-adapted embeddings.
- Improve skill extraction using named entity recognition, ontology-based matching, or synonym/semantic skill matching.
- Use a larger and more diverse resume–job dataset.
- Include experience duration, seniority, education and achievements as additional matching features.
- Build a web interface where users can upload a resume and paste a job description.
- Provide ranked job recommendations or ranked candidate recommendations.
- Evaluate the system with additional metrics and human qualitative assessment.

10. Conclusion

SmartResume demonstrates a practical NLP-based approach to resume–job matching and skill-gap analysis. The project combines a traditional TF-IDF and cosine-similarity baseline with semantic embeddings generated by the all-MiniLM-L6-v2 Sentence Transformer. A hybrid score then combines these two signals, while rule-based skill analysis provides a more interpretable view of the result.

The experiments showed that TF-IDF and semantic similarity behave differently even though they are positively correlated. The controlled evaluation produced ROC-AUC values of 0.853 for TF-IDF and 0.843 for semantic similarity. The aviation case study further demonstrated a large difference between lexical and semantic similarity, supporting the value of comparing both approaches.

The project also highlighted the importance of careful evaluation. Since the dataset's existing AI score is not independently verified, it was not treated as ground truth. Instead, controlled mismatched pairs were used for a transparent experimental comparison. Overall, the project provided practical experience with text preprocessing, TF-IDF, cosine similarity, transformer-based embeddings, skill extraction, evaluation and result visualization.

References

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